

AE 6353 Equation Sheet — Test #1

September 2026

$$F = \frac{Gm_1m_2}{r^2} \quad F_i = G \sum_{\substack{j=1 \\ j \neq i}}^n \frac{m_i m_j}{r_{ij}^3} \bar{\mathbf{r}}_{ij}$$

$$E = T + V \quad T = \frac{1}{2} \sum_{i=1}^n m_i v_i^2 \quad V = -\frac{1}{2} G \sum_{i=1}^n \sum_{\substack{j=1 \\ j \neq i}}^n \frac{m_i m_j}{r_{ij}}$$

$$\ddot{\bar{\mathbf{r}}} = -\frac{\mu}{r^3} \bar{\mathbf{r}}$$

$$\mathcal{E} = \frac{v^2}{2} - \frac{\mu}{r} \quad \bar{\mathbf{h}} = \bar{\mathbf{r}} \times \bar{\mathbf{v}} \quad h = rv \cos \gamma$$

$$r = \frac{p}{1 + e \cos \theta} \quad p = h^2/\mu$$

$$\mu \bar{\mathbf{e}} = \bar{\mathbf{v}} \times \bar{\mathbf{h}} - \frac{\mu}{r} \bar{\mathbf{r}}$$

$$r_p = a(1 - e) \quad r_a = a(1 + e) \quad p = a(1 - e^2)$$

$$e = c/a \quad e = \frac{r_a - r_p}{r_a + r_p}$$

$$\mathcal{E} = -\frac{\mu}{2a} \quad \mathbb{P} = 2\pi \sqrt{a^3/\mu}$$

$$v_c = \sqrt{\mu/r} \quad v_{esc} = \sqrt{2\mu/r}$$

$$\mathbf{v} = \left(-\frac{\mu}{h} \sin \theta\right) \hat{\mathbf{p}} + \left[\frac{\mu}{h}(e + \cos \theta)\right] \hat{\mathbf{q}}$$

$$R = \mu/h \quad \bar{\mathbf{c}}_h = \frac{\mu e}{h} \hat{\mathbf{q}} \quad \tan \beta = \frac{\sin \theta}{e + \cos \theta}$$

$$M = n(t - t_0) = E - e \sin E$$

$$\sin \theta = \frac{\sqrt{1 - e^2} \sin E}{1 - e \cos E} \quad \cos \theta = \frac{\cos E - e}{1 - e \cos E}$$

$$\sin E = \frac{\sqrt{1 - e^2} \sin \theta}{1 + e \cos \theta} \quad \cos E = \frac{\cos \theta + e}{1 + e \cos \theta}$$

$$t - t_0 = \frac{1}{2\sqrt{\mu}} (pD + (1/3)D^3) \quad t - t_0 = \sqrt{\frac{(-a)^3}{\mu}} (e \sinh F - F)$$

$$dx/dt = \sqrt{\mu}/r \quad z = x^2/a$$

$$\sqrt{\mu}\Delta t = x^3S + \frac{\bar{\boldsymbol{r}} \cdot \bar{\boldsymbol{v}}}{\sqrt{\mu}}x^2C + r_1x(1-zS)$$

$$r = \sqrt{\mu} \frac{dt}{dx} = x^2C + \frac{\bar{\boldsymbol{r}} \cdot \bar{\boldsymbol{v}}}{\sqrt{\mu}}x(1-zS) + r_1(1-zC)$$

$$C(z) = \frac{1 - \cos \sqrt{z}}{z} = \frac{1 - \cosh \sqrt{-z}}{z} = \sum_{k=0}^{\infty} \frac{(-z)^k}{(2k+2)!}$$

$$S(z) = \frac{\sqrt{z} - \sin \sqrt{z}}{\sqrt{z^3}} = \frac{\sinh \sqrt{-z} - \sqrt{-z}}{\sqrt{(-z)^3}} = \sum_{k=0}^{\infty} \frac{(-z)^k}{(2k+3)!}$$

$$x = \sqrt{a}\Delta E \quad x = \sqrt{-a}\Delta F$$

For arbitrary vectors $\bar{\boldsymbol{a}}, \bar{\boldsymbol{b}}, \bar{\boldsymbol{c}} \in \mathbb{R}^3$

$$\bar{\boldsymbol{a}} \times (\bar{\boldsymbol{b}} \times \bar{\boldsymbol{c}}) = (\bar{\boldsymbol{a}} \cdot \bar{\boldsymbol{c}})\bar{\boldsymbol{b}} - (\bar{\boldsymbol{a}} \cdot \bar{\boldsymbol{b}})\bar{\boldsymbol{c}}$$

$$\bar{\boldsymbol{a}} \cdot (\bar{\boldsymbol{b}} \times \bar{\boldsymbol{c}}) = (\bar{\boldsymbol{a}} \times \bar{\boldsymbol{b}}) \cdot \bar{\boldsymbol{c}}$$